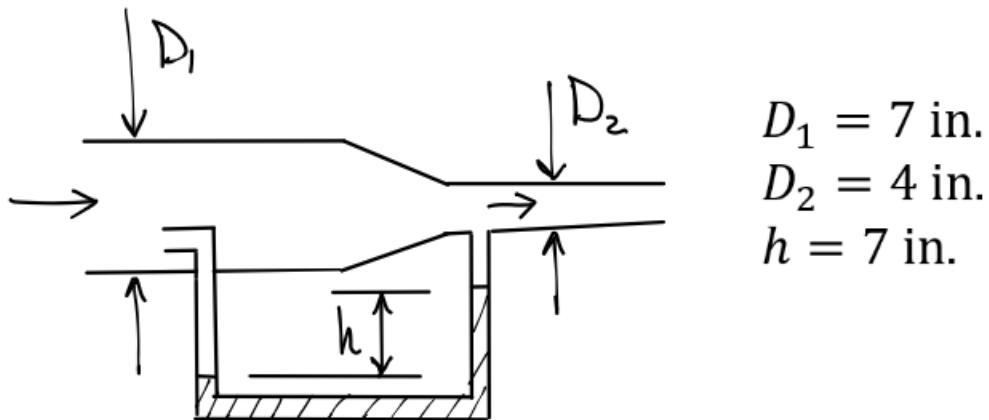


AE 319: Aerodynamics
HW#3

Problem 1

Consider the incompressible flow of air through a circular pipe. A pitot-static probe is used to measure the upstream stagnation pressure at station 1 and the downstream static pressure at station 2. Assuming the density of air to be $0.00238 \text{ slug/ft}^3$, compute the volumetric flow rate in ft^3/s . The fluid in the manometer is an oil with a density of $\rho_{oil} = 1.7704 \text{ slug/ft}^3$. The centerline of the pipe is horizontal, so you may neglect any difference in potential energy. You may also assume that there is no friction.



Problem 2

Consider stagnation point potential flow over a flat plate as shown below on the left. The stream function for this flow is given by

$$\psi = Axy$$

where A is a constant. The origin O here is the stagnation point. (Note - You can use the stream function above to prove this. You may verify this if you desire additional practice). To this stagnation point flow we add a source of strength Λ which generates potential flow over bump as shown below on the right. This would simulate flow inside a heat exchanger, for example, when there is some form of mineral deposition. Another example would be a jet exhaust impinging on an object during take off.

- (a) Determine the bump height h .
- (b) Determine the surface streamline ψ_s .
- (c) Write a script to plot the streamlines. Use $A = \Lambda = 1$. Plot ψ_s using a different color. Highlight the location of the stagnation point.

Hint: ψ_s has to pass through the stagnation point.

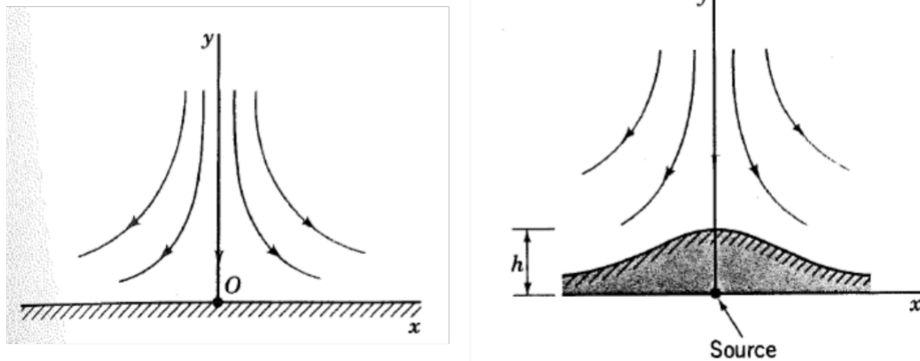


Figure 1: Illustration of the flow over a flat plate and a bump.

Problem 3

A circular cylinder of diameter 0.4 m rotates at 90 rpm in a uniform stream of 10 m/s.

- (a) Locate the stagnation points
- (b) Calculate the rotational speed when the stagnation point moves off the cylinder